

Green Favoritism: Leader Birth Regions Receive Less Pollution-Intensive Growth

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We replicate the birth-region nightlights favoritism documented by Bora (2025) and Düben, Hodler, and Raschky (2026) and extend the result to air pollution. We define *green favoritism* as a negative birth-region coefficient on pollution intensity, defined as $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$. A negative coefficient means birth-region growth is more nightlight-intensive relative to measured air pollution than non-birth-region growth in the same country-year; it is a relative shift in the pollution-to-nightlights ratio, not direct evidence of abatement. Using a harmonized DMSP-to-VIIRS panel for 2005–2017 (622,868 district-year observations), the birth-region coefficient on log nightlights is 0.049 (SE = 0.015). NO₂ intensity falls by 0.039 (SE = 0.019) and PM_{2.5} intensity by 0.044 (SE = 0.015) at leader entry. An event study with 120 balanced spells shows flat NO₂-intensity pre-trends and a negative coefficient becoming statistically significant at $t + 2$ before attenuating by $t + 3$. The effect is substantially stronger in Sub-Saharan Africa, where the nightlights coefficient is roughly three times the non-SSA estimate, although the non-SSA coefficient is itself positive and significant. Globally, the activity effect is also about three times larger in mineral-rent economies, suggestive of a composition channel in which leaders of resource-rich states direct large public works toward home districts; the satellite data cannot, however, distinguish composition from selective enforcement or sectoral sorting.

regional favoritism | political economy | nighttime lights | air pollution
| green favoritism

Political leaders direct resources toward their birth regions. Cross-country panels of subnational nightlights have documented this pattern repeatedly: districts that are home to the sitting national leader grow brighter relative to comparable districts in the same country-year (1–3). A natural follow-up question, however, has not been examined: *is birth-region growth more or less pollution-intensive than activity elsewhere?* If leaders channel public goods and infrastructure toward home districts, pollution should rise less than economic activity. If they route industrial subsidies or manufacturing instead, pollution should rise proportionally or more.

We find that birth-region growth is, in this relative sense, less pollution-intensive. Using a harmonized DMSP-to-VIIRS panel for 2005–2017, we first replicate the nightlights result with a birth-region coefficient of 0.049 (SE = 0.015, $p = 0.001$). Both NO₂ intensity and PM_{2.5} intensity—defined as $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$ —are significantly negative: -0.039 (SE = 0.019) and -0.044 (SE = 0.015). Birth-region growth is more nightlight-intensive relative to measured air pollution than activity in the rest of the same country-year, consistent with a lower pollution-to-nightlights ratio rather than direct evidence of abatement. We term this pattern *green favoritism*.

An event study with a balanced ± 3 -year window around leader entry (120 spells, entry 2008–2016) provides the timing

evidence and the parallel-trends test. Pre-trends for NO₂ intensity at $t = -3$ and $t = -2$ are statistically insignificant; this is consistent with parallel trends for NO₂ intensity, though two insignificant leads in a balanced sample of 120 spells have limited power and are not strong proof. NTL and PM_{2.5}-intensity pre-trends are significant. For NTL, the pre-trend means birth regions were already on a different growth trajectory in the years before leader entry, so the event study does not support a clean causal interpretation for nightlights on its own; for PM_{2.5} intensity, the pre-trend cannot be explained mechanically by activity differences (because intensity already differences out nightlights), so the PM_{2.5} event-study coefficients should be interpreted more cautiously than the NO₂ ones. The NO₂-intensity coefficient becomes negative and statistically significant at $t + 2$ (-0.110^*) and attenuates by $t + 3$.

Hodler and Raschky (2014) establish birth-region favoritism using DMSP nightlights. Bora (2025) and Düben, Hodler, and Raschky (2026) confirm the result with updated birthplace data and extended samples. De Luca et al. (2018) extend the favoritism result to ethnic homelands (4), and Bompreszi et al. (2024) document a parallel pattern for the spousal birthplaces of leaders (5). The literature on regional favoritism is, in this sense, well populated on the side of *how much* activity birth regions receive, but largely silent on the *kind* of activity they

Significance Statement

Birth regions of national leaders grow brighter while in office, and the growth is less pollution-intensive than activity elsewhere. Using a harmonized 2005–2017 panel, we find a nightlights coefficient of 0.049 ($p = 0.001$), an NO₂ intensity coefficient of -0.039 ($p = 0.044$), and a PM_{2.5} intensity coefficient of -0.044 ($p = 0.004$). Birth regions grow brighter without a proportional rise in measured air pollution; this is a relative shift in the pollution-to-nightlights ratio, not direct evidence of abatement. An event study with 120 balanced spells shows flat NO₂-intensity pre-trends and a negative coefficient at $t + 2$. The effect is substantially stronger in Sub-Saharan Africa than elsewhere, and globally the activity coefficient is about three times larger in mineral-rent economies, suggestive of a composition channel in which leaders direct large public works funded by resource rents toward birth regions, though the satellite design cannot separate composition from selective enforcement or sorting.

R.S. conceived the study, built and integrated the main data pipeline, ran the core empirical analysis, and drafted the manuscript. M.M.A. contributed to the NO₂ analysis. R.W. contributed to manuscript review and proofreading.

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receive.

A separate literature studies the political economy of pollution. Cross-country work links environmental outcomes to trade openness, factor endowments, and the strength of regulatory institutions (6). Our paper sits at the intersection of this literature and the favoritism literature: the favoritism literature has examined how much activity birth regions receive but not what kind, while the pollution literature has examined cross-country and within-country drivers of emissions but not the role of leader birthplace. The contribution here is to show that political connection shapes not only the level of activity but its measured pollution-to-nightlights ratio. The pollution-intensity outcome uses independent satellite data on air quality, but it is not a separate identifying margin: under the same regressor and fixed-effects structure the intensity coefficient is mechanically $\beta_{\text{pollution}} - \beta_{\text{NTL}}$, so the substantive value of the intensity outcome is in characterizing the kind of growth birth regions receive rather than in providing a new source of identification.

This paper documents green favoritism, a pollution-intensity pattern characterized using independent satellite data on air quality. The effect is substantially stronger in Sub-Saharan Africa, where the nightlights coefficient is roughly three times the non-SSA estimate, though the non-SSA estimate is itself positive and significant.

Section 1 describes the data. Section 2 presents the replication. Section 3 documents green favoritism. Section 4 turns to the event study. Section 5 examines geographic heterogeneity, and Section 6 discusses mechanisms.

1. Data

The analysis combines five core inputs. First, the Political Leaders' Affiliation Database (PLAD) provides leader identity, spell timing, and subnational birthplace information used to code treatment (7). Second, nightlights provide the economic-activity proxy. For replication we use a DMSP ADM2 panel for 1992–2013. For the main analysis we use a harmonized DMSP-to-VIIRS panel for 2005–2017. DMSP records luminance as an integer Digital Number (DN) from 0 to 63, with known saturation near the ceiling (8). Our zonal statistics pipeline treats zero-valued pixels as valid dark observations rather than missing data, preserving the extensive margin of electrification. Third, we use ACAG surface NO₂ data for 2005–2017 (9). Fourth, we use ACAG SatPM PM_{2.5} as a complementary pollution measure. Fifth, GADM ADM2 boundaries provide the common subnational geography (10).

Raster extraction follows a uniform pipeline across all satellite products. Annual rasters are aggregated to ADM2 polygons by computing the mean of all valid pixels intersecting each boundary polygon (zonal statistics). Zero-valued pixels (districts with no detectable nightlights or surface concentrations) are treated as valid observations rather than missing data, which preserves the extensive margin of electrification and retains variation across dark, low-pollution regions. Only pixels designated `nodata` by the source product (ocean, ice, and out-of-bounds cells) are excluded from the mean. Outcome variables enter regressions as $\log(Y_{ict} + 0.01)$; the constant shift avoids undefined logarithms for zero-valued observations and maps dark districts to approximately -4.6 . Results are robust to alternative shift values of 0.001 and 1. Pollution intensity is $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$, measuring

Table 1. Summary statistics, 2005–2017 analysis sample

Variable	Mean	SD	N
Log nightlights (harmonized)	0.508	2.351	622,868
NO ₂ (ppb)	0.972	2.079	585,141
PM _{2.5} (μg/m ³)	18.051	11.472	620,805
Birth-region indicator	0.003	0.057	622,868

Unit of observation is ADM2 district-year. The birth-region indicator equals one when the district is the birth region of the sitting national leader. Observations differ across variables due to differences in satellite coverage.

emission output per unit of proxied economic activity.

Treatment is defined at the district-year level. A district is treated when it is the birth region of the national leader in office in that year. To ensure a unique treatment assignment in transition years, we truncate the earlier of any two consecutive leader spells that overlap within a country.

The core estimating equation uses ADM2 fixed effects and country-year fixed effects:

$$\log(Y_{ict} + 0.01) = \beta \text{BirthRegion}_{ict} + \alpha_i + \gamma_{ct} + \varepsilon_{ict},$$

where i indexes ADM2 regions, c countries, and t years. The ADM2 fixed effects α_i absorb all time-invariant district characteristics: geography, historical institutions, and long-run income. The country-year fixed effects γ_{ct} absorb national policy shocks, macroeconomic cycles, and any aggregate trend in pollution or nightlights common to all districts of a country in a given year. The coefficient β is therefore identified by within-country-year variation: whether a district that becomes the current leader's birth region experiences differential growth relative to other districts in the same country at the same time.

The main threat to identification is pre-existing trends. Leaders may disproportionately originate from districts already on different growth trajectories, so that the post-treatment levels reflect pre-existing conditions rather than causal favoritism. This is the selection concern raised by Hodler and Raschky (2014), and the event study in Section 4 tests it directly by examining pre-period coefficients. The pollution-intensity outcome carries an additional advantage: because it differences out the nightlights component, common drivers of both nightlights and pollution (including most pre-existing economic trends) cancel. This makes the green favoritism coefficient harder to generate spuriously than the nightlights coefficient alone.

Standard errors are clustered by leader spell in the baseline, following the convention in recent replication work (2). Pollution intensity is $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$; because it uses the same regressor and fixed effects as the component regressions, its coefficient is mechanically $\beta_{\text{pollution}} - \beta_{\text{NTL}}$. A negative intensity coefficient means pollution rises less than nightlights in birth regions.

Table 1 reports summary statistics for the 2005–2017 analysis sample.

2. Replication

Table 2 presents a replication ladder for the birth-region nightlights effect across successive sample windows and sensor generations. In the long DMSP 1992–2013 window the estimate is 0.043 ($p = 0.002$); in the overlap DMSP 2005–2013 window,

Table 2. Replication ladder: birth-region nightlights effect across sensor generations

Step	Outcome	Coef.	SE	<i>p</i>
(1) DMSP 1992–13	Log NTL	0.043	0.014	0.002
(2) DMSP 2005–13	Log NTL	0.033	0.014	0.019
(3) Harm. 2005–17	Log NTL	0.049	0.015	0.001

Lag-0 birth-region coefficients on log nightlights across sensor generations ($N = 1,054,103$; 431,208; 622,868). All rows: ADM2 FE, country-year FE, spell-clustered SE. Row (3) is the specification carried forward to the pollution outcomes in Table 3.

Table 3. Green favoritism: birth regions grow brighter and less pollution-intensive (2005–2017)

	(1) ln NTL	(2) NO ₂ int.	(3) PM _{2.5} int.
Birth-region indicator	0.049*** (0.015)	−0.039* (0.019)	−0.044** (0.015)
<i>N</i>	622,868	585,141	620,805
Region FE	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes
SE clustering	Spell	Spell	Spell

Pollution intensity is $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$. Negative intensity coefficients mean pollution rises less than nightlights in birth regions. Observations differ across columns because NO₂ and PM_{2.5} have different spatial coverage. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Standard errors clustered by leader spell.

0.033 ($p = 0.019$); in the harmonized 2005–2017 panel, 0.049 ($p = 0.001$). The estimates are of similar magnitude and all statistically significant across sensor generations, consistent with Bora (2025) and Düben, Hodler, and Raschky (2026). The harmonized 2005–2017 specification in row (3) is the one we extend to pollution outcomes in Section 3.

3. Green Favoritism

Table 3 presents the main result. In the harmonized 2005–2017 panel, the birth-region coefficient on log nightlights is 0.049 (SE = 0.015, $p = 0.001$), directly replicating the result in Table 2. The new result is the pollution side: the birth-region coefficient on NO₂ intensity is −0.039 (SE = 0.019, $p = 0.044$) and on PM_{2.5} intensity is −0.044 (SE = 0.015, $p = 0.004$). Both pollution-intensity coefficients are negative and statistically significant. Birth regions receive economic activity that does not come with proportional air pollution.

The green favoritism result does not require that pollution falls in absolute terms. The lag-0 NO₂ level coefficient is small and insignificant (0.006, $p = 0.585$); the lag-0 PM_{2.5} level coefficient is similarly small and not significantly different from zero. The negative intensity is driven by a large positive nightlights coefficient combined with a near-zero pollution response: birth regions grow brighter without growing proportionally dirtier. The intensity coefficients reported in Table 3 are estimated directly from the intensity outcome rather than constructed by subtraction; because the level and intensity regressions are run on slightly different effective samples (NO₂ and PM_{2.5} have different spatial coverage from nightlights), the intensity coefficient is not exactly equal to the difference of the level coefficients reported in the text.

The two pollution indicators are complementary rather than interchangeable. The NO₂-intensity estimate is sensi-

Table 4. Green favoritism across lags (2005–2017)

	(1) ln NTL	(2) NO ₂ int.	(3) PM _{2.5} int.
Lag 0	0.049*** (0.015)	−0.039* (0.019)	−0.044** (0.015)
Lag 1	0.063*** (0.015)	−0.081*** (0.019)	−0.059*** (0.015)
Lag 2	0.042** (0.016)	−0.070*** (0.021)	−0.039* (0.016)
<i>N</i>	622,868	585,141	620,805
Region FE	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes
SE clustering	Spell	Spell	Spell

Each cell is a separate regression. Lag k uses the birth-region indicator shifted k years forward. Pollution intensity is $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$. All rows use harmonized 2005–2017 panel, ADM2 fixed effects, country-year fixed effects, and standard errors clustered by leader spell. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

tive to country-level clustering and loses statistical significance under that more conservative scheme (Table 8), while the PM_{2.5}-intensity estimate is more stable. Conversely, the NO₂-intensity event study has the cleaner pre-trends (Section 4), while the PM_{2.5} event-study pre-trends warrant caution. PM_{2.5} captures a broader mix of emission sources including transportation and combustion, while NO₂ is more specific to traffic and industrial combustion.

Table 4 shows that the effects persist across lags. The nightlights effect peaks at lag 1 (+0.063, $p < 0.001$), consistent with a one-year delay between fiscal allocation and measurable luminosity. Both pollution-intensity effects remain negative and statistically significant through lag 2, suggesting some persistence in the lower pollution-to-nightlights ratio in the full-panel lag specification.

4. Event Study

Figure 1 shows the event study for log nightlights, NO₂ intensity, and PM_{2.5} intensity. The sample is a balanced ± 3 -year window around leader entry, restricted to spells entering 2008–2016; the post-window for later entries extends beyond 2017, so this analysis uses the harmonized panel through 2019 solely to preserve balance. This yields 120 balanced spells. The reference period is $t = -1$; the solid red line marks entry ($t = 0$) and the dashed red line marks exit.

The NO₂-intensity panel provides the cleanest event-study evidence. Pre-trend coefficients at $t = -3$ and $t = -2$ are close to zero and neither is statistically significant, consistent with parallel trends for NO₂ intensity, though the balanced sample of 120 spells has limited power. The NO₂-intensity coefficient becomes negative and statistically significant at $t + 2$ (−0.110*) and attenuates by $t + 3$. The NTL panel shows significant pre-trends at $t = -3$ and $t = -2$. Because event-study leads measure differential trends relative to $t = -1$, not differences in baseline levels (which the ADM2 fixed effects already absorb), the NTL pre-trends indicate that birth regions were already on a different growth trajectory before leader entry. The event study therefore does not support a clean causal interpretation for the nightlights outcome on its own; we read the NTL favoritism result as a robust descriptive pattern, and rely on the NO₂-intensity event study for the cleaner timing evidence.

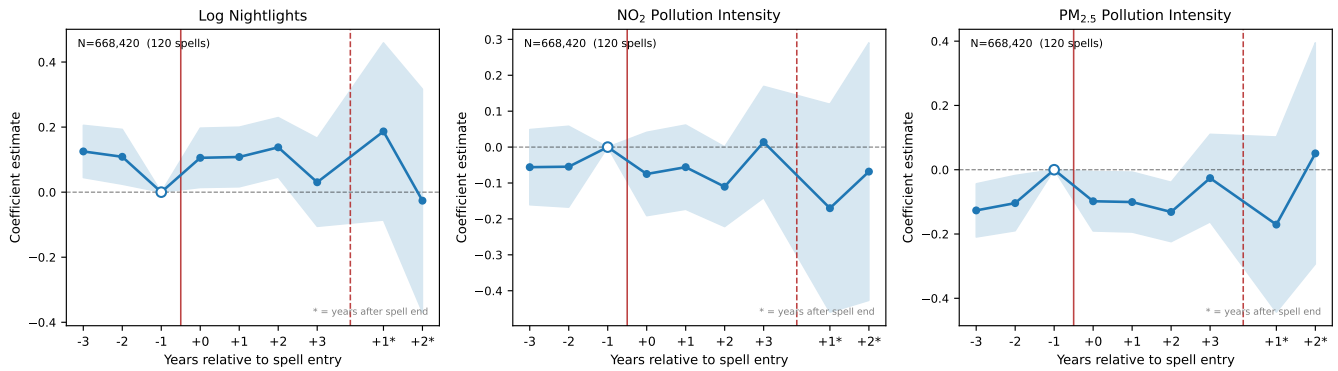


Fig. 1. Event study around leader entry, balanced ± 3 -year window (120 spells, entry 2008–2016). Three panels show log nightlights, NO_2 intensity, and $\text{PM}_{2.5}$ intensity. Coefficients are relative to $t = -1$; solid red line = entry ($t = 0$), dashed red line = exit. Bars are 95% confidence intervals. NO_2 pre-trends at $t = -3$ and $t = -2$ are flat; NTL and $\text{PM}_{2.5}$ pre-trends are significant, consistent with leaders coming from already economically active regions.

The $\text{PM}_{2.5}$ -intensity panel also shows differential pre-trends; this cannot be explained mechanically by activity differences, since intensity differences out the nightlights component, so the $\text{PM}_{2.5}$ event-study estimates should be interpreted more cautiously than the NO_2 ones. $\text{PM}_{2.5}$ intensity remains the more robust outcome under conservative country-level inference (Table 8); the two indicators are complementary rather than interchangeable.

The $t + 3$ fade in all three panels, even with perfectly balanced composition at that horizon, suggests the favoritism effect is short-run rather than a permanent change in district trajectory. Whether this reflects project completion, political cycles, or reduced marginal returns to home-region spending remains an open question.

5. Geographic Heterogeneity

The green favoritism result is substantially stronger in Sub-Saharan Africa than elsewhere, though it is not absent outside SSA. Table 5 decomposes the 2005–2017 nightlights effect by region. The SSA coefficient is 0.109 ($p = 0.017$, $N = 56,589$), roughly three times the non-SSA estimate of 0.033 ($p = 0.026$, $N = 566,279$) and about 2.2 times the pooled estimate of 0.049. The non-SSA coefficient is itself positive and statistically significant; SSA is the more pronounced case rather than the only one. The SSA pollution-intensity estimates are also larger in magnitude: NO_2 intensity at -0.106 (SE = 0.051, $p = 0.042$) and $\text{PM}_{2.5}$ intensity at -0.112 (SE = 0.044, $p = 0.013$). Because satellite measurement error may differ across SSA and non-SSA settings, especially in low-light districts and large ADM2 polygons, the larger SSA coefficients should be interpreted cautiously.

The SSA concentration is consistent across replication windows: the SSA coefficient is 0.125 in DMSP 1992–2013, 0.116 in DMSP 2005–2013, and 0.109 in harmonized 2005–2017, in each case substantially larger than the corresponding non-SSA estimate.

6. Mechanism

The green favoritism pattern is consistent with leaders channeling investment with low emission intensity toward birth

Table 5. Geographic decomposition: 2005–2017 nightlights

Sample	Coef.	SE	p -value	N
Pooled	0.049	0.015	0.001	622,868
Sub-Saharan Africa	0.109	0.046	0.017	56,589
Non-SSA	0.033	0.015	0.026	566,279
Africa	0.101	0.040	0.012	85,063
Low NO_2 dispersion	0.108	0.033	0.001	92,499

Lag-0 birth-region coefficients on log nightlights, harmonized 2005–2017. All rows use ADM2 fixed effects, country-year fixed effects, and leader-spell clustered standard errors. Low- NO_2 -dispersion countries are defined as countries below the median cross-regional standard deviation of NO_2 in 2005–2007.

regions, such as large public works (ports, airports, government facilities) that generate nightlights without proportional industrial emissions. The aggregate satellite design used here cannot directly identify project composition, and the case of Sri Lanka under President Mahinda Rajapaksa (2005–2015) is offered as illustrative rather than as validation of the mechanism. Rajapaksa was born in Hambantota District; during his presidency, the government constructed the Hambantota International Port, Mattala Rajapaksa International Airport, and a major convention center in that district, infrastructure projects that substantially increased local luminosity without industrial activity. Whether this pattern generalizes across countries requires project-level investment data that we do not have.

Three channels could generate negative pollution intensity in birth regions. The first is a *composition channel*: leaders direct low-emission investment (ports, airports, roads, government facilities) toward birth regions instead of heavy manufacturing or industrial plants. The second is an *enforcement channel*: birth-region firms may face stricter environmental enforcement while the leader is in office, reducing emissions per unit of output. The third is a *sorting channel*: birth regions may attract already-clean sectors (services, tourism, knowledge industries) rather than manufacturing. The satellite data used here cannot distinguish these channels, and the democracy split below is consistent with both composition and selective enforcement: low-democracy environments may

Table 6. Mineral-rent decomposition (DMSP 1992–2013)

Sample	Coef.	SE	<i>p</i>
All countries	0.042	0.014	0.003
excl. top-quartile mineral	0.028	0.015	0.061
top-quartile mineral only	0.082	0.034	0.017
Sub-Saharan Africa	0.125	0.031	0.000
excl. top-quartile mineral	0.116	0.041	0.005
top-quartile mineral only	0.139	0.047	0.003
Non-SSA, top-quartile mineral	0.048	0.047	0.314
Non-SSA, excl. top-quartile mineral	0.009	0.016	0.553

Lag-0 birth-region coefficients on log nightlights, DMSP 1992–2013. All rows: ADM2 fixed effects, country-year fixed effects, and leader-spell clustered standard errors. Mineral-rent quartiles are constructed from country-mean WDI mineral rents (% of GDP) over 1992–2013.

permit either more discretionary project allocation or more discretionary enforcement, and the aggregate design cannot tell them apart.

Mineral rents and the composition channel. If the composition channel is correct, the favoritism effect should be stronger in countries with the fiscal slack and political discretion to fund large public-works projects from resource revenues. Large transport and energy infrastructure in resource-rich states (the Hambantota case is one example among many) is precisely the kind of investment that raises nightlights without proportional industrial emissions. We therefore test whether birth-region favoritism varies with exposure to mineral rents.

Table 6 splits the long DMSP 1992–2013 sample by whether a country falls in the top quartile of country-mean mineral rents (% of GDP, World Bank WDI). The long DMSP window is used here because it gives the mineral-rent split adequate statistical power; the same exercise on the harmonized 2005–2017 window points in the same direction but with wider confidence intervals. On the WDI-merged sample, the global activity coefficient is 0.042 ($p = 0.003$). Excluding the top-quartile mineral-rent countries lowers it to 0.028 ($p = 0.061$), while restricting to that quartile alone raises it to 0.082 ($p = 0.017$). Mineral-rent economies thus exhibit a roughly three-fold-larger birth-region nightlights effect than non-mineral economies. Because Table 6 is estimated on the DMSP 1992–2013 nightlights panel rather than the 2005–2017 pollution panel, this evidence is suggestive of stronger *activity* favoritism in mineral-rent economies; whether mineral rents also produce stronger *pollution-intensity* favoritism is less directly tested by this table.

The SSA picture is different. Within SSA, both slices produce large and statistically significant coefficients (0.116 excluding mineral-rent countries, 0.139 in mineral-rent SSA), so the mineral-rent split does not account for the SSA concentration; SSA-specific factors (institutional, demographic, or relating to spell length) carry independent weight. Outside SSA, the activity effect is small and imprecise in both slices. We therefore read the mineral-rent split as one supporting strand of evidence for the composition channel at the global level rather than as the explanation for the SSA pattern. The WDI mineral-rent variable is country-level and slow-moving, so the split absorbs other features of resource-rich states.

Democracy and accountability. Table 7 splits the sample at the median country-mean V-Dem electoral democracy score (0.499)

Table 7. Democracy heterogeneity (2005–2017)

	(1) ln NTL	(2) NO ₂ int.	(3) PM _{2.5} int.
Low democracy	0.066* (0.029)	−0.081* (0.038)	−0.062* (0.029)
High democracy	0.042* (0.018)	−0.021 (0.023)	−0.036* (0.018)
<i>N</i> (low / high)	203k / 408k	185k / 391k	202k / 407k

Sample split at the median country-mean V-Dem electoral democracy score (0.499) over 2005–2017 (11). Pollution intensity is $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$. Each cell is a separate regression with ADM2 fixed effects, country-year fixed effects, and standard errors clustered by leader spell. * $p < 0.05$.

Table 8. Robustness to country-level clustering

Spec.	Coef.	Spell SE	Ctry SE	Ctry <i>p</i>
NL, 1992–2013	0.043	0.014	0.018	0.016
NL, 2005–2013	0.033	0.014	0.017	0.053
NL, 2005–2017	0.049	0.015	0.020	0.013
NO ₂ int., 2005–2017	−0.039	0.019	0.025	0.119
PM _{2.5} int., 2005–2017	−0.044	0.015	0.020	0.025

NL denotes nightlights. Baseline standard errors cluster by leader spell; the robustness column clusters by country, following the original Hodler–Raschky convention.

over 2005–2017. The nightlights favoritism effect is present in both groups, but the green favoritism pattern is concentrated in low-democracy countries. The NO₂ intensity coefficient in high-democracy countries is small and insignificant (-0.021 , $p = 0.35$), while in low-democracy countries it is -0.081 ($p = 0.031$). The PM_{2.5} intensity result is significant in both groups but roughly twice as large in low-democracy contexts.

This pattern is consistent with politically discretionary channels in settings with weaker accountability, including both project composition and selective enforcement. The aggregate satellite design cannot cleanly distinguish these mechanisms; both predict stronger green favoritism where leaders face less monitoring.

Robustness. All main tables cluster standard errors by leader spell, following the convention in recent replication work (2). Table 8 reports the same coefficients with country-level clustering as a more conservative robustness check. The nightlights coefficient in the harmonized window remains significant ($p = 0.013$). The PM_{2.5} intensity result also survives ($p = 0.025$), while NO₂ intensity loses precision ($p = 0.119$). PM_{2.5} thus provides the more conservative evidence for green favoritism under country-level inference.

Limitations. Several caveats apply. First, satellite NO₂ and PM_{2.5} are vertically integrated retrievals translated to surface concentrations by chemical transport models; they are noisy proxies for ground-level exposure and are known to perform less well over bright surfaces (deserts, snow) and over very dark, low-emission regions (9). Classical measurement error in a covariate would attenuate coefficients toward zero, but for log-transformed outcomes the bias direction is not generally signable, so we do not lean on attenuation as a defense of the pollution-intensity result. Second, nightlights are an imperfect proxy for economic activity, especially at low luminance and in agricultural regions where output need not translate

into emitted light. The harmonized DMSP-to-VIIRS series mitigates the saturation issues of raw DMSP (8) but does not eliminate them. Third, birthplace coding in PLAD relies on biographical sources of varying quality; misclassification of a leader's birth region would generally attenuate the treatment indicator. Fourth, the PM_{2.5} result is statistically more robust than the NO₂ result under country-level clustering, and we therefore treat PM_{2.5} as the more conservative evidence for green favoritism while NO₂ provides the cleaner event-study pre-trends; the two indicators are complementary rather than interchangeable. Fifth, the balanced event-study sample of 120 spells is selected by data availability around leadership transitions and is smaller than the main panel, so the event-study estimates should be read as complementary rather than definitive. Sixth, because treatment timing is staggered and leadership spells turn on and off, the two-way fixed-effects estimates may be affected by recent concerns about forbidden comparisons in staggered DiD settings; robustness using modern staggered-adoption estimators (Callaway–Sant'Anna; de Chaisemartin–D'Haultfoeuille) is an important extension we leave for future work. Finally, the paper documents a pattern but cannot isolate the underlying channel. Composition, enforcement, and sorting all remain consistent with the aggregate evidence, and distinguishing them requires project-level or firm-level data not available at the global panel scale used here.

Discussion

This paper asked whether birth-region growth is more or less pollution-intensive than activity elsewhere. We find that it is less so, in the relative sense that the pollution-to-nightlights ratio is lower in birth regions than in comparable districts of the same country-year. Birth-region favoritism in nightlights is a durable feature of the data, present across sensor generations, sample windows, and clustering schemes. The new result is that this favoritism is environmentally selective: birth regions grow brighter without a proportional rise in measured air pollution. Negative pollution-intensity coefficients, flat NO₂-intensity pre-trends in the event study, and the SSA concentration all point to birth-region growth that differs in composition—or in measured emissions per unit of luminosity—from growth elsewhere. The mineral-rent decomposition supplies one supporting strand of evidence at the activity margin: globally, the nightlights effect is about three times larger in mineral-rent economies than outside them, consistent with discretionary resource revenues funding the kind of large, observable public works (ports, airports, roads, government facilities) that the Hambantota case illustrates. Whether mineral rents also amplify the pollution-intensity result is less directly tested, since Table 6 uses the DMSP nightlights panel rather than the pollution panel. The mineral-rent split does not account for the Sub-Saharan concentration: within SSA, the activity effect is large and significant in both mineral-rent and non-mineral slices, so SSA-specific factors carry independent weight. The data do not let us cleanly isolate composition from selective enforcement or sectoral sorting; the aggregate pattern is suggestive of compositional differences in leader birth-region investment but does not by itself identify the channel.

The finding bears on environmental inequality, but only in a relative sense. The estimates suggest a shift in the pollution-

to-nightlights ratio away from leader birth regions; the welfare and exposure implications require population-weighted analysis of pollution levels that we do not undertake here. These results should therefore be read as descriptive evidence of differential pollution-to-nightlights trajectories around leader birth regions, not as definitive evidence of abatement or of a uniquely identified mechanism. Despite these limitations, the paper adds an environmental dimension to the birth-region favoritism literature: political favoritism is visible not only in activity proxies but also in the relationship between activity and satellite-measured air pollution. Future work could test whether specific project categories—port and airport construction, electrification, road building—account for the lower-intensity pattern by linking nightlights and pollution outcomes to administrative investment records, and could revisit the staggered DiD inference using modern estimators robust to forbidden comparisons.

Materials and Methods

Data sources and panel construction are described in Section 1. The nightlights panels are built at ADM2 resolution by extracting annual zonal means from DMSP DN rasters (1992–2013) and harmonized DMSP-to-VIIRS rasters (2005–2017) using GADM level-2 boundaries. NO₂ and PM_{2.5} are extracted from ACAG annual surface-concentration grids using the same boundaries. Leader birthplace districts are coded from PLAD and matched to GADM ADM2 polygons. The democracy split uses country-mean V-Dem electoral democracy scores (v2x_polyarchy) over 2005–2017 (11), with countries split at the cross-country median (0.499). All regressions use `linearmodels` PanelOLS with absorbed fixed effects; standard errors are clustered by leader spell unless otherwise noted.

Data, Materials, and Code Availability

Code and manuscript sources are available at <https://github.com/RichSchulz/political-leader-pollution>. External inputs include PLAD (7), GADM (10), ACAG NO₂ (9) and PM_{2.5}, Li et al. harmonized nightlights, and DMSP annual composites obtained under the providers' access conditions.

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